

Unit 12 – Evaluation of Unit 6 Assignment vs Unit 11 Assignment

In unit 6, I was placed in a group of six and tasked with simulating a development team in Airbnb to apply regression to determine which variables influence the price of a listing in New York. We also employed K-means clustering to create groups of listings depending on different markets in New York, such as budget and location pairings.

My unit 11 assignment consisted of developing a recommender system using a publicly available Spotify dataset. By this time I was more confident in my coding abilities which meant I could develop the system testing two models: a Random Forest (RF) regression model and a multi-layer perceptron (MLP) model. It was my first opportunity of the module to develop a model end-to-end, from loading the data, to completing EDA to understand its characteristics, engineering of custom features, model identification and development, evaluation, and explainability using Shapely Additive Explanations (SHAP). Once developed, I had to create a recorded presentation discussing the dataset, model, evaluation, and explainability, while also considering the ethics behind the data collected and how the model would be deployed into the wider technological architecture.

I found completing the unit 11 assignment to be more difficult overall as I could not rely on other ideas from my teammates like I did during the unit 6 assignment. I was had to identify which models to develop and how they should be evaluated and explained. Not being able to ask my course mates which models they would develop meant that I had to trust my own understanding of machine ML principles. I ultimately received a strong grade for my assignment, but it's likely that reviewing certain decisions with them, such as the choice of dataset, would have contributed to a higher grade.